An Exact Odd-Level Metaplectic Family for a Toral Hénon Map and a Growing No-Action-Alias Window

Hénon Route-A Working Series, C131

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Abstract

For the toral Hénon automorphism generated by $A = \left(\begin{smallmatrix} 3 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, we extend a single level-seven quantization to every odd integer $N \geq 3$. One frozen half-phase, Fourier, and chirp convention gives an explicit unitary on $\mathbb{C}^N$, exact Egorov covariance for all N^2 Weyl observables, the unchanged discrete clock, and a uniform antiunitary reversal. An exact integer recurrence also gives a growing interval in which reduction modulo N cannot alias the classical action to the identity. Nine prime and composite certificate levels check 25,313 Weyl cases. The theorem is per-level: it asserts neither cross-level projective compatibility nor a semiclassical trace match.

1 Progress beyond one finite level

An exact lift at one modulus can be a phase accident. The advance here is a single theorem quantified over every odd modulus, including composite ones, with the same signs and reversor. A second theorem makes the pre-alias range explicit and arbitrarily long as the level grows. Neither statement compares finite quantum traces with an external spectrum.

2 Frozen family

Let $N \geq 3$ be odd, write $\omega_N = e^{2\pi i/N}$, and let $h_N \in \mathbb{Z}/N\mathbb{Z}$ satisfy $2h_N = 1$. On the standard basis of $\mathcal{H}_N = \mathbb{C}^N$, set

$$Q_N|x\rangle = \omega_N^x|x\rangle, \qquad P_N|x\rangle = |x+1\rangle,$$

$$W_N(q,p) = \omega_N^{-h_N qp} Q_N^q P_N^p, \quad (\mathcal{F}_N)_{xy} = N^{-1/2}\omega_N^{xy},$$

$$(C_N)_{xx} = \omega_N^{3h_N x^2}, \qquad U_N = C_N \mathcal{F}_N^{-1}.$$

All residues in operator exponents are modulo N.

Theorem 1 (all-odd exact lift)**.** *For every odd $N \geq 3$, U_N is unitary and*

$$U_N W_N(q,p) U_N^{-1} = W_N(3q-p, q)$$

for all $(q,p) \in (\mathbb{Z}/N\mathbb{Z})^2$. Hence U_N^n implements A^n after exactly n steps. If K is coefficientwise conjugation and $\Theta_N = \mathcal{F}_N K$, then

$$\Theta_N^2 = I, \qquad \Theta_N U_N \Theta_N^{-1} = U_N^{-1}, \qquad \Theta_N W_N(q,p)\Theta_N^{-1} = W_N(p,q).$$

Proof. Finite Fourier orthogonality proves that $\mathcal{F}_N$ is unitary, and C_N is diagonal unitary. The Weyl product is

$$W_N(v)W_N(w) = \omega_N^{h_N(v_1w_2-v_2w_1)}W_N(v+w).$$

Direct conjugation gives $U_NQ_NU_N^{-1} = W_N(3,1)$ and $U_NP_NU_N^{-1} = W_N(-1,0)$. In a general $W_N(q,p)$, its prefactor $\omega_N^{-h_Nqp}$ cancels the product phase $\omega_N^{h_Nqp}$ from $W_N(3q,q)W_N(-p,0)$, proving Egorov without a scalar repair.

Since $\overline{\mathcal{F}_N} = \mathcal{F}_N^{-1}$ and $\overline{C_N} = C_N^{-1}$, $(\mathcal{F}_NK)^2 = I$ and

$$\mathcal{F}_N\overline{U_N}\mathcal{F}_N^{-1} = \mathcal{F}_NC_N^{-1} = U_N^{-1}.$$

Finally $KW_N(q,p)K = W_N(-q,p)$; Fourier conjugation and one Weyl reordering give $W_N(p,q)$. □

3 A growing no-action-alias window

Define $u_0 = 0$, $u_1 = 1$, and $u_{n+1} = 3u_n - u_{n-1}$.

Proposition 1 (integer pre-alias certificate)**.** *For every $n \geq 1$,*

$$A^n = \begin{pmatrix} u_{n+1} & -u_n \\ u_n & -u_{n-1} \end{pmatrix}, \qquad \|A^n - I\|_{\max} = u_{n+1} - 1.$$

If $N > u_{L+1} - 1$, then $A^n \not\equiv I \pmod N$ and U_N^n is not scalar for every $1 \leq n \leq L$.

Proof. The matrix formula follows by induction from the recurrence. Positivity and monotonicity show that the upper-left entry of $A^n - I$ has maximal absolute value. If its max norm is below N, congruence to zero modulo N would make every entry literally zero, contradicting $u_n > 0$. A scalar U_N^n has trivial adjoint action; exact Egorov and injectivity of the Weyl labels would then force $A^n \equiv I \pmod N$. □

At $N = 3, 9, 23, 57, 145$, the certified windows are respectively $1, 2, 3, 4, 5$. This shows growth, but not a trace asymptotic.

4 Exact certificate and controls

The receipt covers levels $3, 5, 7, 9, 11, 15, 23, 57, 145$, including composites, and binds 25,313 Egorov cases plus the corresponding antiunitary Weyl cases by case-ledger hashes. An independent implementation reconstructs every digest, threshold, residue, convention, and nonclaim. SymPy checks 50,670 exact identities; twenty-six repaired-hash mutations attack phase signs, reversors, windows, labels, and scope.

At even N, $2h = 1 \pmod N$ has no solution, so this literal $qp/2$ convention cannot be copied unchanged. This is not a no-go theorem for doubled-phase or other separately defined even-level Weil/metaplectic conventions. Likewise, the per-level theorem supplies no preferred relative phase between levels.

5 Route-A boundary

The named Hilbert spaces, exact unitaries, observable covariance, clock, and antiunitary justify

$$(\text{A1\_WEAK}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_NATURAL\_QUANTIZATION}).$$

There is no target divisor census, analytic target completion, cross-level projective theorem, or semiclassical trace match. Route B is unauthorized, and the active scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

6 Conclusion

C131 replaces an isolated finite lift by an exact all-odd family and supplies a growing interval free of adjoint-action aliasing. The next separate task would need a rigorously normalized cross-level or semiclassical structure; it is not inferred here.